

WASP-36b

R-1420

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220210 · created 2026-08-01T06:12:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459620.7901 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.048
Transit depth (Rp/Rs)^2	2.44 +/- 1.5 [%]
Orbital Inclination (inc)	87.94 +/- 2.9
Airmass coefficient 1 (a1)	0.67 +/- 0.43
Airmass coefficient 2 (a2)	0.3 +/- 1.27
Scatter in the residuals of the lightcurve fit is	69.78 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.0925 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.048 vs 0.1368 (deviation 0.3σ)
Duration fit vs published	0.0925 d vs 0.0758 d (deviation 1.45σ)
Mid-time O–C	-8.9 min
Depth SNR	2.4
Residual scatter	69.78 %

Light curve

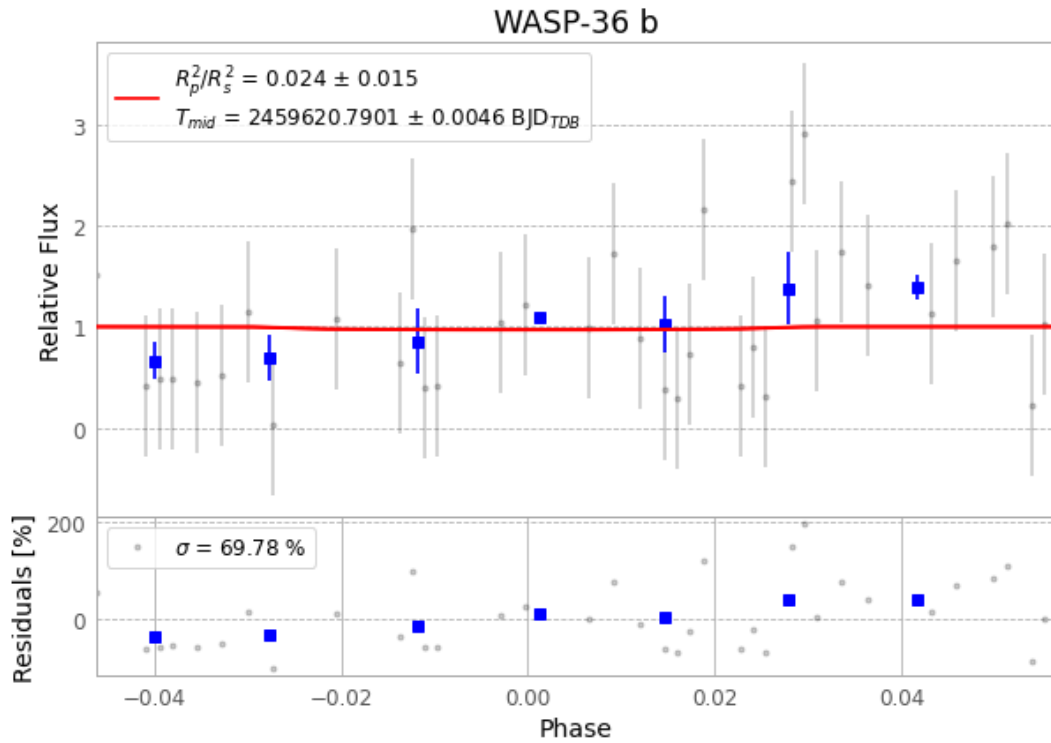


Figure 1 — FinalLightCurve_WASP-36 b_2022-02-09.png (SHA-256 f4b04c6bfa2aa9b6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 421], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 630918f2fec85b6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36220210050612.FITS → WASP-36220210085412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220210.log (d9ed843433de...), FinalLightCurve_WASP-36 b_2022-02-09.png (f4b04c6bfa2aa9b6...), FinalLightCurve_WASP-36 b_2022-02-09.csv (0caaf23ccd6b...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:12Z.